

Review on Interference and Diffraction

Phys2500J

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Overview

1. Problem Set

2. Diffraction

Mirage

Question

In the desert, it is hot on the ground surface. As a result, the hot air near ground has smaller index than cooler air higher up. Assume the index has the relation with the height above the ground which is $n(y) = 1 + \alpha y$. Find the trajectory for the light rays which has an initial angle of 30° at the height of h

Approximation for Young's double

Two way of shifting

Anti-reflective coating

Question

Why the surface of your cellphone's camera lens looks colored? e.g. especially looks blue or purple.

Anti-reflective coating

Diffraction

Definition

Light ray tends to bend around objects. This tendency is called diffraction. It is a property of waves.

Examples

How to use Huygens's principle to analyze diffraction

- Fresnel diffraction: Source, screen, and obstacle are close together.
- Fraunhofer diffraction: Source, screen, and obstacle are far apart.

Fraunhofer Diffraction

Equation

Fraunhofer diffraction:

$$\tilde{U}(\theta) = C \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} T(x, y) e^{-iks \sin \theta (x \cos \varphi + y \sin \varphi)} dx dy$$

Examples

Single slit:

$$I = I_0 \left(\frac{\sin(\frac{\beta}{2})}{\frac{\beta}{2}} \right)^2$$

$$\beta = \frac{2\pi}{\lambda} a \sin \theta$$

$$\sin \theta = \frac{m\lambda}{a} \quad m\text{th dark fringes}$$

Fraunhofer Diffraction

Several Slits

Equation

$$\begin{aligned}\tilde{U}(\theta) &= \int_{\Sigma} U_0(x) e^{ikr} dx & r_j &= L_j - x_j \sin \theta \\ &= \sum e^{ikL_j} \int_{\Sigma_j} U_0(x_j) e^{-ikx_j \sin \theta} dx_j & L_j &= L_1 + j \cdot (d \sin \theta) \\ &= \tilde{N}(\theta) \tilde{u}(\theta)\end{aligned}$$

Where

$$\begin{aligned}\tilde{N}(\theta) &= C_N \frac{\sin N \frac{\delta}{2}}{\sin \frac{\delta}{2}} & \delta &= \frac{2\pi d \sin \theta}{\lambda} \\ \tilde{u}(\theta) &= C_u \frac{\sin \frac{\phi}{2}}{\frac{\phi}{2}} & \phi &= \frac{2\pi a \sin \theta}{\lambda}\end{aligned}$$

Several Slits

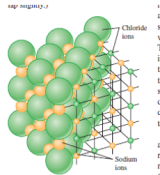
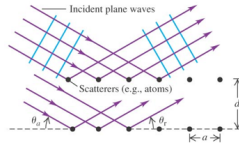
Circular Apertures

Relation with Fourier Transform

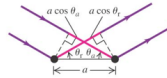
X-ray diffraction

- The *Bragg condition* for constructive interference is

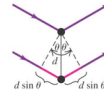
(a) Scattering of waves from a rectangular array



(b) Scattering from adjacent atoms in a row
Interference from adjacent atoms in a row is constructive when the path lengths $a \cos \theta_i$ and $a \cos \theta_r$ are equal, so that the angle of incidence θ_i equals the angle of reflection (scattering) θ_r .



(c) Scattering from atoms in adjacent rows
Interference from atoms in adjacent rows is constructive when the path difference $2d \sin \theta$ is an integral number of wavelengths, as in Eq. (36.16).



$$2d \sin \theta = m\lambda.$$

The End